

```
root@eltex-practice2-pg1-v14:~# su - user1
user1@eltex-practice2-pg1-v14:~$ ps axo pid,nlwp | awk '$2 > 1' | wc -l
11
user1@eltex-practice2-pg1-v14:~$
ps axo pid,comm,nlwp | awk '$3 > 1' | head -15
  PID COMMAND      NLWP
  724 unattended-upgr  2
 14206 qemu-ga        2
 15306 polkitd         4
 15318 ModemManager   4
 25284 multipathd       7
 25906 rsyslogd          4
 36961 systemd-timesyn  2
 36963 upowerd           4
 38863 udisksd            6
 62430 fwupd             6
```

```
217.71.138.1 - PuTTY
Fields Management for window 1:Def, whose current sort field is %CPU
Navigate with Up/Dn, Right selects for move then <Enter> or Left commits,
'd' or <Space> toggles display, 's' sets sort. Use 'q' or <Esc> to end!

* PID      = Process Id      PGRP      = Process Grou OOMS      = OOMEM Score  RSS      = Res Mem (sma
* USER     = Effective Us   TTY      = Controlling ENVIRON   = Environment  PSS      = Proportion R
* PR       = Priority       TPGID     = Tty Process vMj      = Major Faults PSan     = Proportion A
* NI       = Nice Value     SID      = Session Id  vMn      = Minor Faults PSfd     = Proportion F
* VIRT     = Virtual Imag  nTH      = Number of Th USED     = Res+Swap Siz PSsh     = Proportion S
* RES      = Resident Siz  P        = Last Used Cp nsIPC     = IPC namespac USS      = Unique RSS,
* SHR      = Shared Memor  TIME     = CPU Time  nsMNT     = MNT namespac ioR      = I/O Bytes Re
* S        = Process Stat  SWAP     = Swapped Size nsNET     = NET namespac ioRop    = I/O Read Ope
* %CPU     = CPU Usage     CODE     = Code Size (K nsPID     = PID namespac ioW      = I/O Bytes Wr
* TIME+    = CPU Time, hu  DATA    = Data+Stack ( nsUSER    = USER namespa ioWop    = I/O Write Op
* COMMAND  = Command Name nMaj     = Major Page F nsUTS     = UTS namespac AGID     = Autogroup Id
* PPID     = Parent Proce nMin     = Minor Page F LXC     = LXC containe AGNI     = Autogroup Ni
* %MEM     = Memory Usage  nDRT     = Dirty Pages RSan     = RES Anonymou STARTED  = Start Time f
* UID      = Effective Us  WCHAN    = Sleeping in RSfd     = RES File-bas ELAPSED  = Elapsed Runn
* RUID     = Real User Id  Flags    = Task Flags < RSlk     = RES Locked ( %CUU    = CPU Utilizat
* RUSER    = Real User Na CGROUPS   = Control Grou RSsh     = RES Shared ( %CUC    = Utilization
* SUID     = Saved User I  SUPGIDS  = Supp Groups CGNAME    = Control Grou nsCGROUP = CGRP namespa
* SUSER    = Saved User N  SUPGRPS  = Supp Groups NU      = Last Used NU nsTIME   = TIME namespa
* GID      = Group Id     TGID     = Thread Group LOGID     = Login User I
* GROUP    = Group Name   OOMa     = OOMEM Adjust EXE      = Executable P
```

```
217.71.138.1 - PuTTY
top - 12:08:17 up 7 days, 3:41, 1 user, load average: 0.00, 0.01, 0.00
Tasks: 108 total, 1 running, 107 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni, 98.0 id, 2.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 3916.0 total, 1144.2 free, 494.1 used, 2581.8 buff/cache
MiB Swap: 3185.0 total, 3185.0 free, 0.0 used. 3421.9 avail Mem

  PID USER      PR  NI  S   %CPU   TIME+  COMMAND                                %MEM  RUSER
 15306 polkitd   20   0  S    0.1  0:08.64 polkitd                                0.2  polkitd
 67995 userl      20   0  R    0.1  0:00.02 top                                    0.2  userl
   1 root       20   0  S    0.0  0:16.35 systemd                                0.3  root
   2 root       20   0  S    0.0  0:00.08 kthreadd                                0.0  root
   3 root       20   0  S    0.0  0:00.00 pool_workqueue_release                 0.0  root
   4 root       0 -20  I    0.0  0:00.00 kworker/R-rcu_g                        0.0  root
   5 root       0 -20  I    0.0  0:00.00 kworker/R-rcu_p                        0.0  root
   6 root       0 -20  I    0.0  0:00.00 kworker/R-slub_                        0.0  root
   7 root       0 -20  I    0.0  0:00.00 kworker/R-netns                        0.0  root
  12 root       0 -20  I    0.0  0:00.00 kworker/R-mm_pe                        0.0  root
  13 root       20   0  I    0.0  0:00.00 rcu_tasks_kthread                      0.0  root
  14 root       20   0  I    0.0  0:00.00 rcu_tasks_rude_kthread                 0.0  root
  15 root       20   0  I    0.0  0:00.00 rcu_tasks_trace_kthread                0.0  root
  16 root       20   0  S    0.0  0:00.24 ksoftirqd/0                            0.0  root
  17 root       20   0  I    0.0  0:05.73 rcu_preempt                            0.0  root
  18 root       rt   0  S    0.0  0:02.17 migration/0                           0.0  root
  19 root      -51   0  S    0.0  0:00.00 idle_inject/0                          0.0  root
```

217.71.138.1 - PuTTY

```
root@eltex-practice2-pgl-v14:~# passwd
New password: 
```

217.71.138.1 - PuTTY

```
top - 12:10:44 up 7 days, 3:43, 2 users, load average: 0.00, 0.00, 0.00
Tasks: 111 total, 1 running, 110 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.2 sy, 0.0 ni, 99.7 id, 0.2 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 3916.0 total, 1139.5 free, 498.9 used, 2581.8 buff/cache
MiB Swap: 3185.0 total, 3185.0 free, 0.0 used. 3417.1 avail Mem
```

PID	RUSER	USER	PR	NI	S	%CPU	%MEM	TIME+	COMMAND
1	root	root	20	0	S	0.0	0.3	0:16.36	systemd
2	root	root	20	0	S	0.0	0.0	0:00.08	kthreadd
3	root	root	20	0	S	0.0	0.0	0:00.00	pool_workqueue_relea+
4	root	root	0	-20	I	0.0	0.0	0:00.00	kworker/R-rcu_g
5	root	root	0	-20	I	0.0	0.0	0:00.00	kworker/R-rcu_p
6	root	root	0	-20	I	0.0	0.0	0:00.00	kworker/R-slub
7	root	root	0	-20	I	0.0	0.0	0:00.00	kworker/R-netns
12	root	root	0	-20	I	0.0	0.0	0:00.00	kworker/R-mm_pe
13	root	root	20	0	I	0.0	0.0	0:00.00	rcu_tasks_kthread
14	root	root	20	0	I	0.0	0.0	0:00.00	rcu_tasks_rude_kthre+
15	root	root	20	0	I	0.0	0.0	0:00.00	rcu_tasks_trace_kthr+
16	root	root	20	0	S	0.0	0.0	0:00.24	ksoftirqd/0
17	root	root	20	0	I	0.0	0.0	0:05.73	rcu_preempt
18	root	root	rt	0	S	0.0	0.0	0:02.17	migration/0
19	root	root	-51	0	S	0.0	0.0	0:00.00	idle_inject/0
20	root	root	20	0	S	0.0	0.0	0:00.00	cpuhp/0
21	root	root	20	0	S	0.0	0.0	0:00.00	cpuhp/1

217.71.138.1 - PuTTY

```
top - 12:12:30 up 7 days, 3:45, 2 users, load average: 0.00, 0.00, 0.00
Tasks: 113 total, 1 running, 112 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni, 100.0 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 3916.0 total, 1139.5 free, 498.9 used, 2581.8 buff/cache
MiB Swap: 3185.0 total, 3185.0 free, 0.0 used. 3417.1 avail Mem
Send pid 68143 signal [15/sigterm] 15
```

PID	RUSER	USER	PR	NI	S	%CPU	%MEM	TIME+	COMMAND
68143	user1	user1	20	0	S	0.0	0.1	0:00.01	bash

```
217.71.138.1 - PuTTY
top - 12:13:12 up 7 days, 3:46, 2 users, load average: 0.00, 0.00, 0.00
Tasks: 113 total, 1 running, 112 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.1 us, 0.1 sy, 0.0 ni, 99.8 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 3916.0 total, 1139.5 free, 498.9 used, 2581.8 buff/cache
MiB Swap: 3185.0 total, 3185.0 free, 0.0 used. 3417.1 avail Mem
Send pid 68143 signal [15/sigterm] 2
  PID RUSER  USER    PR NI S   %CPU %MEM    TIME+  COMMAND
  68143 user1    user1    20  0 S    0.0  0.1    0:00.01 bash
```

```
%Cpu(s): 0.0 us, 0.1 sy, 0.0 ni, 99.9 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 3916.0 total, 1139.5 free, 498.9 used, 2581.8 buff/cache
MiB Swap: 3185.0 total, 3185.0 free, 0.0 used. 3417.1 avail Mem
Send pid 68143 signal [15/sigterm] 3
  PID RUSER  USER    PR NI S   %CPU %MEM    TIME+  COMMAND
  68143 user1    user1    20  0 S    0.0  0.1    0:00.01 bash
```

```
Tasks: 113 total, 1 running, 112 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni, 100.0 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 3916.0 total, 1139.5 free, 498.9 used, 2581.8 buff/cache
MiB Swap: 3185.0 total, 3185.0 free, 0.0 used. 3417.1 avail Mem
Send pid 68143 signal [15/sigterm] 9
  PID RUSER  USER    PR NI S   %CPU %MEM    TIME+  COMMAND
  68143 user1    user1    20  0 S    0.0  0.1    0:00.01 bash
```

```
217.71.138.1 - PuTTY
user1@eltex-practice2-pgl-v14:~$ passwd
Changing password for user1.
Current password: Killed
root@eltex-practice2-pgl-v14:~#
```

```
217.71.138.1 - PuTTY
top - 12:13:53 up 7 days, 3:46, 2 users, load average: 0.00, 0.00, 0.00
Tasks: 111 total, 1 running, 110 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni, 99.8 id, 0.0 wa, 0.0 hi, 0.0 si, 0.2 st
MiB Mem : 3916.0 total, 1139.9 free, 498.4 used, 2581.8 buff/cache
MiB Swap: 3185.0 total, 3185.0 free, 0.0 used. 3417.6 avail Mem

  PID RUSER   USER      PR  NI  S   %CPU  %MEM    TIME+  COMMAND
  
```

```
~/file_task3.txt" [New]      ^Z      0,0-1      All

[1]+  Stopped                  vim ~/file_task3.txt
root@eltex-practice2-pgl-v14:~# sleep 600
^Z
[2]+  Stopped                  sleep 600
root@eltex-practice2-pgl-v14:~# jobs
[1]-  Stopped                  vim ~/file_task3.txt
[2]+  Stopped                  sleep 600
root@eltex-practice2-pgl-v14:~# bg %2
[2]+  sleep 600 &
root@eltex-practice2-pgl-v14:~# pgrep sleep
68228
root@eltex-practice2-pgl-v14:~# renice 10 $(pgrep sleep)
68228 (process ID) old priority 0, new priority 10
root@eltex-practice2-pgl-v14:~# ps axl | grep sleep
0      0      68228   68213  30  10      5684   2048 do_sys SN   pts/0      0:00 sleep 600
0      0      68236   68213  20  0       6544   2304 pipe_r S+   pts/0      0:00 grep --color=auto sleep
root@eltex-practice2-pgl-v14:~#
```

```
217.71.138.1 - PuTTY
root@eltex-practice2-pgl-vl4:~# kill -15 %2
root@eltex-practice2-pgl-vl4:~# jobs
[2]+  Terminated                  sleep 600
root@eltex-practice2-pgl-vl4:~#
```

```
217.71.138.1 - PuTTY
root@eltex-practice2-pgl-vl4:~# trap 'echo "Меня голыми руками не возьмёшь!"' SIGINT SIGQUIT
root@eltex-practice2-pgl-vl4:~# kill -SIGINT $$
Меня голыми руками не возьмёшь!
root@eltex-practice2-pgl-vl4:~# kill -SIGQUIT $$
Меня голыми руками не возьмёшь!
root@eltex-practice2-pgl-vl4:~# trap - SIGINT SIGQUIT
root@eltex-practice2-pgl-vl4:~#
```

РАЗДЕЛ 2

```
217.71.138.1 - PuTTY
root@eltex-practice2-pgl-vl4:~# nohup ~/cuckoo.sh &
[1] 68249
root@eltex-practice2-pgl-vl4:~# nohup: ignoring input and appending output to 'nohup.out'

root@eltex-practice2-pgl-vl4:~# pgrep -a cuckoo
68249 /bin/bash /root/cuckoo.sh
root@eltex-practice2-pgl-vl4:~# cat ~/cuckoo.log
2026-03-19 12:17:16 Startup!
root@eltex-practice2-pgl-vl4:~#
```

```
217.71.138.1 - PuTTY
GNU nano 7.2 /tmp/crontab.jaMRs3/crontab *
#
# To define the time you can provide concrete values for
# minute (m), hour (h), day of month (dom), month (mon),
# and day of week (dow) or use '*' in these fields (for 'any').
#
# Notice that tasks will be started based on the cron's system
# daemon's notion of time and timezones.
#
# Output of the crontab jobs (including errors) is sent through
# email to the user the crontab file belongs to (unless redirected).
#
# For example, you can run a backup of all your user accounts
# at 5 a.m every week with:
# 0 5 * * 1 tar -zcf /var/backups/home.tgz /home/
#
# For more information see the manual pages of crontab(5) and cron(8)
#
# m h dom mon dow   command
*/1 * * * * /bin/bash /root/observer.sh
```

^G Help	^O Write Out	^W Where Is	^K Cut	^T Execute	^C Location	M-U Undo
^X Exit	^R Read File	^_ Replace	^U Paste	^J Justify	^/ Go To Line	M-E Redo

```
# daemon's notion of time and timezones.
#
# Output of the crontab jobs (including errors) is sent through
# email to the user the crontab file belongs to (unless redirected)
#
# For example, you can run a backup of all your user accounts
# at 5 a.m every week with:
# 0 5 * * 1 tar -zcf /var/backups/home.tgz /home/
#
# For more information see the manual pages of crontab(5) and cron(
#
# m h dom mon dow   command

*/1 * * * * /bin/bash /root/observer.sh
root@eltex-practice2-pgl-vl4:~#
```

217.71.138.1 - PuTTY

```
root@eltex-practice2-pgl-vl4:~# ln -s ~/template_task.sh ~/task_backup
root@eltex-practice2-pgl-vl4:~# ln -s ~/template_task.sh ~/task_monitor
root@eltex-practice2-pgl-vl4:~# ln -s ~/template_task.sh ~/task_cleanup
root@eltex-practice2-pgl-vl4:~# cat > ~/observer.conf << 'EOF'
/root/cuckoo.sh
/root/task_backup
/root/task_monitor
/root/task_cleanup
EOF
root@eltex-practice2-pgl-vl4:~# cat ~/observer.conf
/root/cuckoo.sh
/root/task_backup
/root/task_monitor
/root/task_cleanup
root@eltex-practice2-pgl-vl4:~# ls -la ~/task_*
lrwxrwxrwx 1 root root 22 Mar 19 12:19 /root/task_backup -> /root/template_task.sh
lrwxrwxrwx 1 root root 22 Mar 19 12:19 /root/task_cleanup -> /root/template_task.sh
lrwxrwxrwx 1 root root 22 Mar 19 12:19 /root/task_monitor -> /root/template_task.sh
root@eltex-practice2-pgl-vl4:~#
```

217.71.138.1 - PuTTY

```
root@eltex-practice2-pgl-vl4:~# sleep 120
root@eltex-practice2-pgl-vl4:~# cat ~/cuckoo.log
2026-03-19 12:17:16 Startup!
2026-03-19 12:21:02 task_backup[68742] 6
2026-03-19 12:21:02 task_monitor[68984] 8
2026-03-19 12:21:02 task_cleanup[69230] 7
2026-03-19 12:22:01 task_backup[69678] 8
2026-03-19 12:22:01 task_monitor[69920] 5
2026-03-19 12:22:01 task_cleanup[70166] 6
2026-03-19 12:23:01 task_backup[70612] 10
2026-03-19 12:23:01 task_monitor[70852] 4
2026-03-19 12:23:01 task_cleanup[71096] 8
2026-03-19 12:24:01 task_backup[71542] 4
2026-03-19 12:24:01 task_monitor[71782] 5
2026-03-19 12:24:02 task_cleanup[72026] 8
root@eltex-practice2-pgl-vl4:~# cat ~/observer.log
2026-03-19 12:21:02 Restarting: task_backup
2026-03-19 12:21:02 Restarting: task_monitor
2026-03-19 12:21:02 Restarting: task_cleanup
2026-03-19 12:22:01 Restarting: task_backup
2026-03-19 12:22:01 Restarting: task_monitor
2026-03-19 12:22:01 Restarting: task_cleanup
2026-03-19 12:23:01 Restarting: task_backup
2026-03-19 12:23:01 Restarting: task_monitor
2026-03-19 12:23:01 Restarting: task_cleanup
2026-03-19 12:24:01 Restarting: task_backup
2026-03-19 12:24:01 Restarting: task_monitor
2026-03-19 12:24:02 Restarting: task_cleanup
root@eltex-practice2-pgl-vl4:~#
```

217.71.138.1 - PuTTY

```
root@eltex-practice2-pgl-vl4:~# ls ~/report_*.log 2>/dev/null || echo "Нет report файлов ещё"
Нет report файлов ещё
root@eltex-practice2-pgl-vl4:~#
```



```
217.71.138.1 - PuTTY
root@eltex-practice2-pgl-v14:~# journalctl -u cron | tail -10
Mar 19 12:21:02 eltex-practice2-pgl-v14 CRON[68320]: pam_unix(cron:session): session closed for user root
Mar 19 12:22:01 eltex-practice2-pgl-v14 CRON[69256]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
Mar 19 12:22:01 eltex-practice2-pgl-v14 CRON[69257]: (root) CMD (/bin/bash /root/observer.sh)
Mar 19 12:22:01 eltex-practice2-pgl-v14 CRON[69256]: pam_unix(cron:session): session closed for user root
Mar 19 12:23:01 eltex-practice2-pgl-v14 CRON[70192]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
Mar 19 12:23:01 eltex-practice2-pgl-v14 CRON[70193]: (root) CMD (/bin/bash /root/observer.sh)
Mar 19 12:23:01 eltex-practice2-pgl-v14 CRON[70192]: pam_unix(cron:session): session closed for user root
Mar 19 12:24:01 eltex-practice2-pgl-v14 CRON[71122]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
Mar 19 12:24:01 eltex-practice2-pgl-v14 CRON[71123]: (root) CMD (/bin/bash /root/observer.sh)
Mar 19 12:24:02 eltex-practice2-pgl-v14 CRON[71122]: pam_unix(cron:session): session closed for user root
root@eltex-practice2-pgl-v14:~#
```

217.71.138.1 - PuTTY

```
root@eltex-practice2-pgl-vl4:~# kill -SIGUSR1 $(pgrep -f cuckoo.sh)
root@eltex-practice2-pgl-vl4:~# crontab -l | grep -v observer.sh | crontab -
[1]+  Done                  nohup ~/cuckoo.sh
root@eltex-practice2-pgl-vl4:~# crontab -l
# Edit this file to introduce tasks to be run by cron.
#
# Each task to run has to be defined through a single line
# indicating with different fields when the task will be run
# and what command to run for the task
#
# To define the time you can provide concrete values for
# minute (m), hour (h), day of month (dom), month (mon),
# and day of week (dow) or use '*' in these fields (for 'any').
#
# Notice that tasks will be started based on the cron's system
# daemon's notion of time and timezones.
#
# Output of the crontab jobs (including errors) is sent through
# email to the user the crontab file belongs to (unless redirected).
#
# For example, you can run a backup of all your user accounts
# at 5 a.m every week with:
# 0 5 * * 1 tar -zcf /var/backups/home.tgz /home/
#
# For more information see the manual pages of crontab(5) and cron(8)
#
# m h  dom mon dow   command
root@eltex-practice2-pgl-vl4:~# cd ~
root@eltex-practice2-pgl-vl4:~# tar -czf task3_report.tar.gz \
  cuckoo.log \
  observer.log \
  cuckoo.sh \
  template_task.sh \
  observer.sh \
  observer.conf \
  $(ls report_*.log 2>/dev/null)
root@eltex-practice2-pgl-vl4:~#
```

217.71.138.1 - PuTTY

```
root@eltex-practice2-pgl-vl4:~# ls -lh ~/task3_report.tar.gz
-rw-r--r-- 1 root root 1.8K Mar 19 12:25 /root/task3_report.tar.gz
root@eltex-practice2-pgl-vl4:~# tar -tzf ~/task3_report.tar.gz
cuckoo.log
observer.log
cuckoo.sh
template_task.sh
observer.sh
observer.conf
root@eltex-practice2-pgl-vl4:~#
```